Regime-Switching Dynamics and Tail Dependence

A Multi-Stage Statistical Framework for Gold's Safe-Haven Properties

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Abstract

This paper proposes a multi-stage statistical framework to investigate the conditional safe-haven properties of gold relative to U.S. equities over the period 2006–2026. Using a GARCH-X baseline to quantify volatility spillovers, quantile regression to characterize distributional heterogeneity, and a GARCH-HMM regime-detection pipeline as an alternative to VIX threshold rules, we examine whether gold's negative correlation with equities is state-dependent rather than unconditional. The framework is validated against four historical tail events and compared on the basis of regime onset lead time and false-positive rate during non-crisis periods.

Keywords: safe haven, regime switching, GARCH-HMM, quantile regression, tail dependence, gold.

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1 Introduction

Gold is widely regarded as a safe-haven asset, yet empirical evidence reveals a more complex picture. During the 2022 Federal Reserve tightening cycle, gold and equities declined simultaneously; in early 2020, gold fell alongside equities during the initial COVID-19 panic before recovering. These episodes suggest that gold's protective properties are state-dependent rather than unconditional — an asset may hedge under normal conditions without providing crisis protection when it is needed most.

This paper addresses two questions: whether gold's safe-haven properties are activated conditionally on market regimes, and whether a data-driven regime classifier can identify stress onset earlier and more reliably than conventional VIX threshold rules. To this end, we propose a multi-stage framework combining a GARCH-X volatility baseline, quantile regression for tail dependence analysis, and a Gaussian Hidden Markov Model (HMM) for latent regime detection (Hamilton, 1989). We evaluate the framework against four historical stress episodes and compare HMM-based regime identification with the VIX > 30 threshold on the basis of lead time and false-positive rate.

Research Gap

Although extensive evidence documents gold's role as a hedge and safe-haven asset (see Appendix A for a full review), several important gaps remain.

First, many studies rely on ex-post, static definitions of market stress, such as fixed VIX thresholds or predefined crisis periods. While intuitive, such approaches are subject to arbitrariness and may fail to capture the timing of regime shifts accurately (Hannich et al., 2018).

Second, there is limited research comparing market-based indicators (such as VIX) with data-driven regime detection methods (such as HMM), leading to inconsistencies in how crisis regimes are identified.

Third, although tail behavior is widely recognized as central to safe-haven analysis, relatively few studies integrate tail dependence methods (e.g., quantile regression) with regime-switching frameworks (Chernozhukov, 2005), limiting the ability to explain when and how safe-haven properties are activated.

To address these limitations, this study proposes a multi-stage statistical framework that integrates GARCH volatility modeling, quantile regression, and Hidden Markov Models. This framework enables a systematic evaluation of gold's safe-haven properties and allows for a direct comparison between VIX-based and data-driven regime identification methods in extreme market conditions.

2 Methodology

2.1 Exploratory Analysis

2.1.1 Data Description and Sample Construction

This study uses daily data from 2006 to 2026, covering four major stress episodes: the 2008 Global Financial Crisis, the 2011 Eurozone crisis, the 2020 COVID-19 shock, and the 2022 geopolitical

tightening cycle.

Asset prices are sourced from two providers. Commodity prices — gold (`GC=F`), silver (`SI=F`), platinum (`PL=F`), copper (`HG=F`), and palladium (`PA=F`) — are downloaded as front-month futures via yfinance, alongside the S&P 500 (`^GSPC`) and VIX (`^VIX`). Macroeconomic variables — the 10-year TIPS yield (`DFII10`) and 10-year breakeven inflation (`T10YIE`) — are obtained from FRED.

Returns are computed as log differences $r_t = \ln P_t - \ln P_{t-1}$. VIX and TIPS yields are retained in levels, as differencing would remove economically meaningful information about the prevailing stress and rate environment. For the HMM feature set, EGARCH(1,1) conditional volatilities are estimated for gold and S&P 500 log returns, and gold-to-copper ratios are constructed from the raw price series; all six HMM inputs are subsequently Z-score standardized prior to estimation.

2.1.2 Descriptive Statistics and VIX-Based Regime Analysis

Descriptive Statistics To explore how asset behavior varies across market conditions, we classify observations into three regimes based on VIX levels: *Calm*, *Elevated*, and *Panic*.

The threshold of VIX > 30 is commonly used in the literature and market practice as an indicator of extreme stress (Carr, 2019). While this threshold provides an intuitive starting point, it is inherently arbitrary and will be formally evaluated and refined in later modeling stages (Hannich et al., 2018).

Descriptive statistics—including mean, variance, skewness, and kurtosis—are computed for each asset within each regime. This stratification allows us to identify how return distributions change under different levels of market stress.

Empirical Findings Regime-stratified descriptive statistics (Table 6, Appendix B) reveal that gold is the only asset whose annualized return *increases* under extreme stress (VIX > 30), while the S&P 500 and industrial metals collapse. Persistent excess kurtosis across all regimes confirms fat-tailed behavior that motivates the quantile regression framework in Section 2.3.

The scatter and QQ plots (Figures 7 and 8, Appendix B) show that gold's correlation with the S&P 500 remains near zero regardless of VIX regime — unique among the metals examined — while return distributions deviate substantially from normality under stress, with heavier tails in the extreme regime.

The rolling correlation analysis (Figure 9, Appendix B) reveals that gold's safe-haven activation is episode-specific rather than uniform: correlation dynamics differ markedly across the GFC, COVID shock, and the 2022 tightening cycle, motivating the dynamic regime-switching approach in Section 2.4.

Maximum drawdown comparisons (Figure 10, Appendix B) confirm that gold consistently contains peak-to-trough losses relative to equities and industrial metals across all nine stress episodes, with faster recovery in each case.

Key Insight and Motivation for Further Modeling The VIX-based regime classification provides an initial framework for distinguishing different market conditions and reveals that gold exhibits heterogeneous behavior across regimes.

However, this approach has two fundamental limitations. First, the choice of threshold is arbitrary and may not accurately reflect true regime boundaries. Second, the VIX signal is inherently lagging, as it reflects market expectations rather than real-time structural changes (Carr, 2019).

These limitations motivate the need for more advanced, data-driven modeling approaches. In particular, they provide the foundation for introducing GARCH-based volatility modeling, quantile regression for tail dependence, and Hidden Markov Models for dynamic regime identification.

2.2 Baseline Volatility Model: GARCH-X Framework

2.2.1 Model Setup

The exploratory evidence suggests that gold's risk is not constant across time, motivating a time-varying volatility baseline. The GARCH-X framework extends a standard GARCH specification with an exogenous equity-stress variable, allowing a direct test of whether stock-market turbulence spills over into gold's conditional variance beyond gold's own internal dynamics:

$$r_{\text{gold},t} = \mu + \varepsilon_t \tag{1}$$

$$\varepsilon_t = \sigma_t \, z_t \tag{2}$$

$$\sigma_t^2 = \omega + \alpha \, \varepsilon_{t-1}^2 + \beta \, \sigma_{t-1}^2 + \gamma \, x_{t-1} \tag{3}$$

The coefficient γ is the quantity of primary interest: it measures whether lagged equity-market stress enters gold's variance equation after controlling for gold's own shock and persistence terms. Two specifications of x_{t-1} are estimated. The *symmetric* version uses the absolute value of lagged S&P 500 returns, testing whether large equity moves in either direction raise gold volatility. The *asymmetric* version enters lagged positive and negative returns separately, allowing the model to distinguish downside from upside transmission:

$$\sigma_t^2 = \omega + \alpha \, \varepsilon_{t-1}^2 + \beta \, \sigma_{t-1}^2 + \gamma \left| r_{\text{sp500},t-1} \right| \tag{4}$$

$$\sigma_t^2 = \omega + \alpha \, \varepsilon_{t-1}^2 + \beta \, \sigma_{t-1}^2 + \gamma_1 \, r_{\text{sp500},t-1}^{+} + \gamma_2 \, r_{\text{sp500},t-1}^{-} \tag{5}$$

2.2.2 Empirical Results

Baseline GARCH Parameters Table 9 (Appendix C) reports parameter estimates for both specifications. The results confirm standard features of financial volatility: $\hat{\alpha}$ is positive and highly significant, indicating that gold's own shocks elevate subsequent variance, while $\hat{\beta} \approx 0.957$ reflects strong volatility persistence. The significant low-$\hat{\nu}$ estimate supports the use of a heavy-tailed Student-t distribution over the Gaussian.

Spillover Tests Table 10 (Appendix C) reports the γ estimates. In the symmetric specification, $\hat{\gamma} = 0.380$ ($p \approx 0$), indicating that large equity moves systematically raise gold's conditional variance. The asymmetric results show both $\hat{\gamma}_1 = 0.401$ and $\hat{\gamma}_2 = 0.364$ are highly significant, with the difference between them small and unlikely to be economically meaningful. Equity-market intensity therefore transmits into gold volatility broadly, without a distinctly stronger downside

channel. This implies that gold is not volatility-insulated from systemic stress, even if its return behavior may still differ from equities during crises.

Diagnostics The conditional volatility series (Figure 1) rises sharply around the GFC (2008), COVID-19 (2020), and the 2022 war-and-inflation episode, consistent with the crisis structure identified in the exploratory stage. The distributional gap report (Table 11, Appendix C) reveals a key limitation: although the model successfully filters volatility clustering, residual kurtosis remains at 6.74 and the Jarque–Bera test rejects normality at any conventional level ($p \approx 0$). The standardized residuals remain heavy-tailed, indicating that GARCH-X does not fully capture gold's extreme return behavior, as further illustrated in Figure 11 (Appendix C).

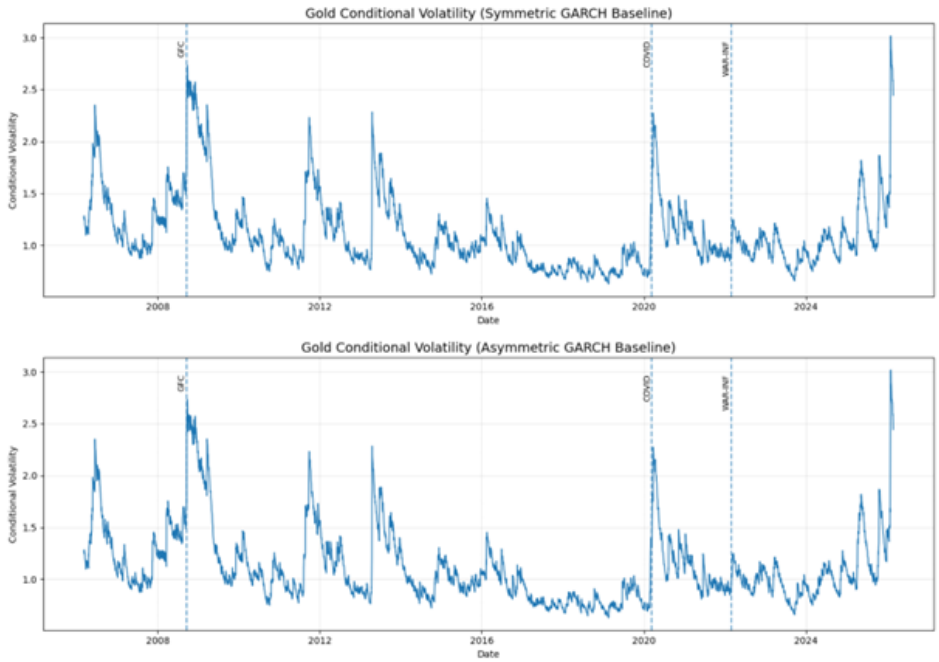

Figure 1: Conditional volatility plots for the symmetric and asymmetric GARCH baselines.

Summary The GARCH-X baseline establishes three points relevant to the subsequent modeling. First, gold volatility is strongly persistent ($\hat{\beta} \approx 0.957$) and responds significantly to equity-market stress, ruling out the view that gold operates as a fully insulated asset. Second, spillover is symmetric across positive and negative equity moves, suggesting that market intensity rather than market direction drives gold's variance response. Third, the persistent heavy tails in standardized residuals indicate that conditional variance alone is insufficient to characterize gold's risk, motivating the regime-switching framework developed in the next section.

2.3 Quantile Regression Analysis

2.3.1 Motivation

Prior sections establish that gold's volatility responds to equity stress but leave its return behavior across market conditions uncharacterised. We employ quantile regression (QR), which esti-

mates the conditional relationship between gold returns and equity/fear indicators separately at each quantile $\tau \in (0, 1)$ allowing the safe-haven hypothesis to be tested where it is most relevant: in the distributional tails.Throughout this section, quantile τ refers to the conditional distribution of gold returns: low τ captures periods when gold itself underperforms, while high τ captures periods of strong gold appreciation. This is distinct from, but often correlated with, the broader market stress regime identified in Section 2.4

2.3.2 Model Specification

For each quantile level $\tau \in (0, 1)$, we estimate the following linear quantile regression model for gold log returns:

$$Q_\tau(r_{\text{gold},t} \mid \mathbf{x}_t) = \alpha(\tau) + \beta_{\text{sp500}}(\tau)\, r_{\text{sp500},t} + \beta_{\text{vix}}(\tau)\, \text{VIX}_t + \varepsilon_t(\tau) \tag{6}$$

where $Q_\tau(\cdot \mid \mathbf{x}_t)$ denotes the τ-th conditional quantile of gold returns given covariates $\mathbf{x}_t$, and $\varepsilon_t(\tau)$ is the quantile-specific error term satisfying $\mathbb{P}(\varepsilon_t(\tau) \leq 0 \mid \mathbf{x}_t) = \tau$. Parameters are estimated by minimizing the Koenker and Bassett (1978) check function:

$$\min_{\alpha,\boldsymbol{\beta}} \sum_{t=1}^{T} \rho_\tau\big(r_{\text{gold},t} - \alpha(\tau) - \beta_{\text{sp500}}(\tau)\, r_{\text{sp500},t} - \beta_{\text{vix}}(\tau)\, \text{VIX}_t\big) \tag{7}$$

where $\rho_\tau(u) = u\,(\tau - \mathbf{1}_{u<0})$ is the asymmetric absolute loss function. We estimate seven quantile levels $\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}$, covering the left tail, median, and right tail of the gold return distribution. Standard errors are obtained via the Koenker and Bassett (1978) rank-inversion method, and 95% confidence intervals are reported for both slope coefficients.

2.3.3 Empirical Results

Gold Is Less Sensitive to Equities in the Lower Tail At the 5th percentile quantile, the coefficient on S&P 500 returns is approximately 0.0099, which is substantially lower than the median estimate of 0.0266. This indicates that when gold experiences weak returns, its co-movement with equities becomes smaller. In stressed environments, gold behaves more independently from the stock market, which supports its safe-haven role.The full quantile process for $\hat{\beta}_{\text{sp500}}$ across $\tau \in \{0.05, 0.10, \ldots, 0.95\}$ is reported in Table 12 (Appendix D).

Market Fear Has an Asymmetric Effect The VIX coefficient increases significantly across quantiles, from -0.0426 in the lower tail to 0.0555 in the upper tail. This suggests that rising market fear does not simply increase gold volatility. Instead, higher fear levels are associated with stronger positive tail returns for gold. In other words, panic conditions may trigger flight-to-safety demand for gold.

Better VaR Performance At the 5% Value-at-Risk level, the QR-VaR exceedance rate is 5.03%, compared to a GARCH-X VaR exceedance rate of 5.40%. Since the theoretical target is 5%, Quantile Regression provides more accurate downside risk estimation than GARCH-X.

Full coefficient estimates and confidence intervals across all quantile levels are reported in Table 12 (Appendix D).

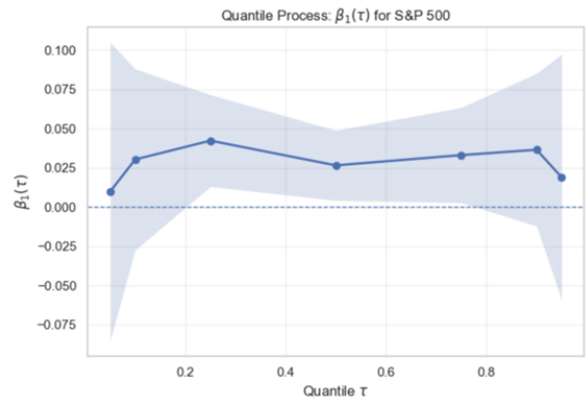

Figure 2: Quantile process for S&P 500 returns.

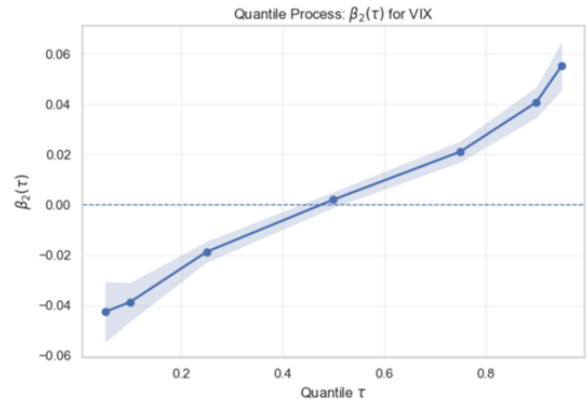

Figure 3: Quantile process for VIX.

Interpretation The results demonstrate that the relationship between gold and financial markets is not constant across return states. Gold's defensive value is mainly concentrated in tail events rather than in normal market periods. Specifically:

- Gold becomes less connected with equities during downside states;
- Gold benefits more from rising fear during upside tail states;
- Tail-based modeling improves risk forecasting accuracy.

Therefore, average regression models may underestimate the true safe-haven property of gold.

Table 1: Left, Median, and Right Tail Comparison

Coefficient	Left Tail ($\tau = 0.05$)	Median ($\tau = 0.50$)	Right Tail ($\tau = 0.95$)	Right $-$ Left
$\hat{\beta}_{\text{sp500}}$	0.0099	0.0266	0.0190	0.0091
$\hat{\beta}_{\text{vix}}$	-0.0426	0.0021	0.0555	0.0981

$\hat{\beta}_{\text{sp500}}$ and $\hat{\beta}_{\text{vix}}$: quantile regression slope coefficients on S&P 500 returns and VIX, respectively. Right $-$ Left denotes the difference between the right-tail ($\tau = 0.95$) and left-tail ($\tau = 0.05$) estimates, capturing asymmetry across the return distribution.

Conclusion Overall, Quantile Regression provides strong evidence of distributional heterogeneity in gold returns. The safe-haven role of gold is activated primarily during extreme market conditions rather than during ordinary periods. Gold is less equity-sensitive in the lower tail and more positively affected by fear shocks in the upper tail.

QR characterizes gold's distributional properties conditional on tail realizations, but does not provide a real-time probabilistic classification of the current market regime. This naturally motivates the next section, where a Hidden Markov Model is used to dynamically classify panic and normal regimes.

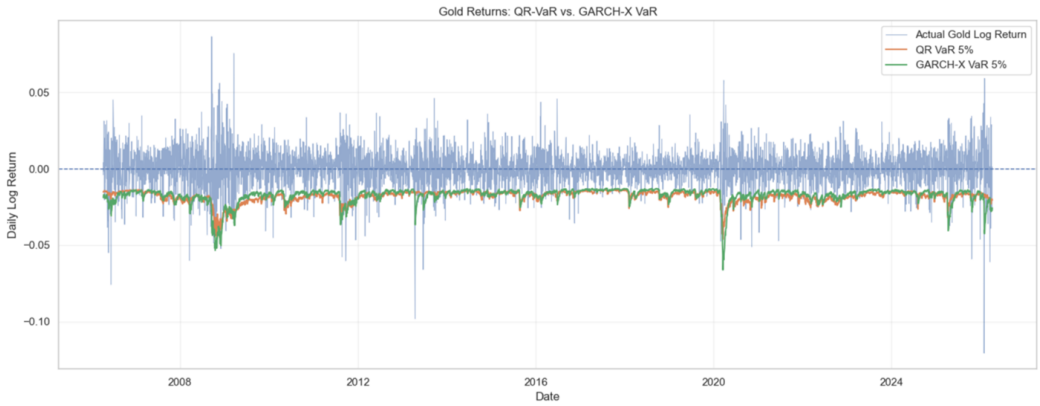

Figure 4: QR-VaR vs. GARCH-X VaR.

Table 2: VaR Backtesting Results

Model	Expected Exceedance	Actual Exceedance	No. Exceedances	Observations
QR-VaR	0.0500	0.0503	252	5014
GARCH-X VaR	0.0500	0.0540	271	5014

Expected exceedance rate is set at the 5% VaR level. A model is better calibrated when its actual exceedance rate is closer to 0.05. QR-VaR denotes the Value-at-Risk estimated via Quantile Regression at $\tau = 0.05$; GARCH-X VaR is estimated from the conditional variance of the GARCH-X model.

2.4 Regime Detection: GARCH-HMM

2.4.1 Motivation: From Static Snapshots to Dynamic Identification

While the Quantile Regression analysis in Section 2.3 characterizes gold's return distribution conditional on tail realizations, it does not provide a real-time probabilistic classification of the current market regime. In financial markets, the true state of the economy is a latent variable: investors observe only price fluctuations and volatility spikes. We therefore implement a two-state Gaussian Hidden Markov Model (HMM) to dynamically identify regime onset, treating market states as discrete, persistent processes that govern the joint dynamics of gold and equities.

2.4.2 Feature Engineering and Economic Rationale

The performance of an HMM is fundamentally dependent on the quality of its "observations." To ensure the model captures the structural shifts in gold's behavior, we select four categories of features:

• **Volatility State Factors (EGARCH Volatility):** We utilize conditional variances extracted from EGARCH(1,1) models for both Gold and the S&P 500. Unlike simple standard deviations, EGARCH captures the leverage effect—the empirical reality that negative shocks in the equity market trigger more significant volatility expansions than positive shocks of the same magnitude. This provides a high signal-to-noise ratio for state detection.

• **Macroeconomic Cost Factors:** We include the 10-Year Real Interest Rate (DFII10), representing the opportunity cost of holding non-yielding gold forming the "fundamental floor" of gold

prices.

• **Commodity Cycle Ratios:** Gold-to-Copper ratios are included to capture "fear-based" outperformance. Historically, gold appreciates relative to industrial commodities during recessionary pivots or geopolitical escalations.

• **Direct Returns:** Logarithmic returns of both assets provide immediate cross-sectional linkage information.

Note on Pre-processing: All features undergo Z-score standardization. This is critical to ensure that variables with different magnitudes (e.g., a 2% return vs. a 30.0 VIX level) contribute proportionately to the likelihood estimation.

2.4.3 Model Specification

Let $\mathbf{o}_t \in \mathbb{R}^6$ denote the standardized observation vector at time t, comprising the six engineered features described above. We model the joint dynamics of $\{\mathbf{o}_t\}_{t=1}^T$ as a two-state Gaussian Hidden Markov Model (HMM) with a discrete latent state $s_t \in \{0, 1\}$ governed by three sets of parameters:

1. **Initial distribution**: $\boldsymbol{\pi} = (\pi_0, \pi_1)$, where $\pi_s = \mathbb{P}(s_1 = s)$.
2. **Transition matrix**: $A \in [0, 1]^{2\times 2}$, with $A_{ij} = \mathbb{P}(s_{t+1} = j \mid s_t = i)$ and $\sum_j A_{ij} = 1$.
3. **Emission distributions**: each state emits observations from a multivariate Gaussian,

$$\mathbf{o}_t \mid s_t = s \;\sim\; \mathcal{N}(\boldsymbol{\mu}_s,\, \Sigma_s)\,, \quad s \in \{0, 1\}, \tag{8}$$

   where $\boldsymbol{\mu}_s \in \mathbb{R}^6$ and $\Sigma_s \in \mathbb{R}^{6\times 6}$ is a full (unrestricted) covariance matrix.

Prior to estimation, all six features are Z-score standardized using their full-sample means and standard deviations, ensuring that variables of differing magnitudes contribute proportionately to the likelihood. The parameter set $\theta = \{\boldsymbol{\pi}, A, \{\boldsymbol{\mu}_s, \Sigma_s\}_{s=0}^1\}$ is estimated by maximizing the observed-data log-likelihood via the Baum–Welch algorithm (Baum et al., 1970), an Expectation–Maximization (EM) procedure that iterates until convergence or a maximum of 1,000 iterations. Once $\hat{\theta}$ is obtained, the most probable state sequence $\hat{\mathrm{s}} = (\hat{s}_1, \ldots, \hat{s}_T)$ is decoded using the Viterbi algorithm (Viterbi, 1967).

To quantify each feature's contribution to regime separation, we define a discriminative score for feature k as

$$D_k = \frac{|\hat{\mu}_{k,1} - \hat{\mu}_{k,0}|}{\hat{\sigma}_{k,0} + \hat{\sigma}_{k,1}}, \tag{9}$$

where $\hat{\mu}_{k,s}$ and $\hat{\sigma}_{k,s}$ denote the estimated state-s mean and marginal standard deviation of feature k from the HMM emission parameters. Higher values of D_k indicate greater regime-discriminating power.

2.4.4 Empirical Results and Model Validation

3.4.4.1 Covariance Structure and Risk Dynamics The transition from State 0 to State 1 is marked by a profound structural transformation in the system's risk profile, as evidenced by the estimated covariance matrices $\hat{\Sigma}_0$ and $\hat{\Sigma}_1$ (Table 7, Appendix B). Table 3 summarizes the three most economically meaningful contrasts.

Table 3: Comparison of Selected Covariance Elements

Feature Pair (Index)	**State 0 (Normal)**	**State 1 (Crisis)**	**Ratio** (Σ_1/Σ_0)
Var(SP500 Log-Ret) [2,2]	0.3294	2.3350	7.09×
Var(SP500 Vol) [6,6]	0.1478	1.6157	10.93×
Cov(Gold Vol, SP500 Vol) [5,6]	0.0286	0.4437	15.51×

The estimated covariance matrices reveal three economically meaningful contrasts between the two regimes. First, diagonal entries — the marginal variances of each feature — expand dramatically in State 1, signalling a broad explosion in risk levels. The variance of S&P 500 log returns, for instance, rises from 0.329 in State 0 to 2.335 in State 1, while the variance of S&P 500 EGARCH conditional volatility increases from 0.148 to 1.616 — an expansion of more than tenfold. This near-uniform scaling of all diagonal elements statistically confirms the characterisation of State 1 as a stress regime marked by extreme dispersion and heightened uncertainty across the entire feature set. Second, the off-diagonal covariances, which remain small in State 0 and reflect meaningful diversification across assets and risk factors, grow substantially in absolute magnitude under State 1. This "correlation breakdown" pattern — in which previously uncorrelated risk drivers begin to co-move — is a well-documented feature of financial crises and renders conventional portfolio diversification ineffective precisely when protection is most needed. Third, and most directly relevant to the safe-haven analysis, the covariance between the two EGARCH volatility features (gold and S&P 500) rises from 0.029 in State 0 to 0.444 in State 1, a fifteenfold increase. This sharp transition from near-independence to strong co-movement constitutes statistical evidence of volatility spillover and risk contagion: equity market turbulence propagates rapidly into gold market uncertainty under the stress regime. It is precisely this shift — from decoupled to synchronized risk dynamics — that the GARCH-HMM framework exploits as the key signal for identifying safe-haven activation.

3.4.4.2 Feature Discriminative Power To validate the feature engineering process, a discriminative score was calculated for each input variable to quantify its contribution to regime separation. As reported in Table 5 (Appendix B), volatility-based features are the primary drivers of state identification: `sp500_egarch_vol` and `gold_egarch_vol` attain scores of 0.7742 and 0.7458, respectively, far exceeding those of the raw log-return series (0.0656 for S&P 500 and 0.0019 for gold). This disparity justifies the feature engineering choice of substituting EGARCH-derived conditional variances for noise-heavy price returns, as the former provide a substantially cleaner signal for the hidden Markov states.

3.4.4.3 Statistical Characterization of Regimes The Gaussian HMM successfully bifurcates the market into two statistically distinct environments. In-sample conditional means and volatilities are reported in Table 8 (Appendix B); the key regime contrast is summarized below.

Notably, while S&P 500 mean returns turn negative in State 1 (-0.078% daily), gold's mean return remains positive ($+0.041\%$), confirming safe-haven activation in the identified stress regime. The concurrent near-doubling of S&P 500 volatility (from 0.777% to 1.504%) corroborates the structural nature of the state separation.

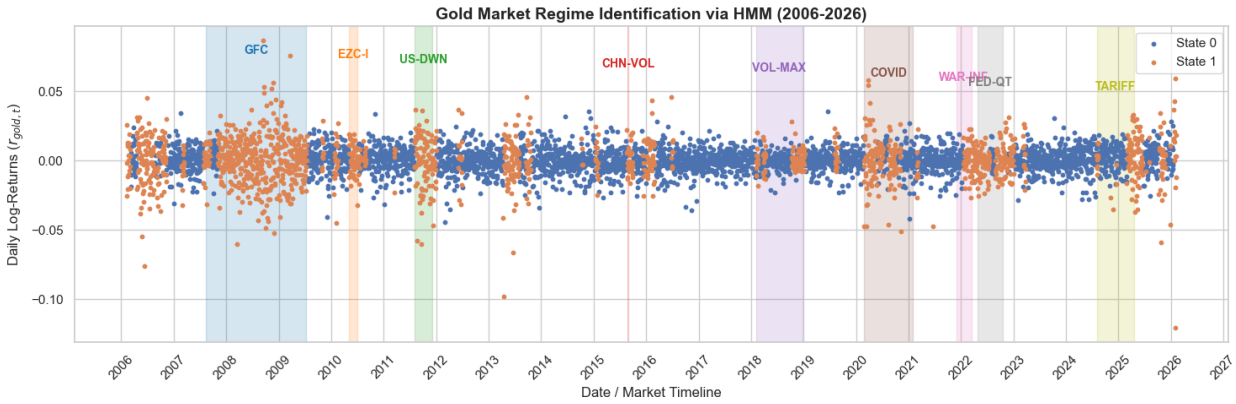

Figure 5: Gold Market Regime Identification via HMM (2006-2026)

3.4.4.4 Regime Path Analysis and Historical Events The decoded state path demonstrates high alignment with major macroeconomic disruptions over the 2006–2026 period. The model's regime classifications align closely with known historical stress episodes. During the Global Financial Crisis, a sustained State 1 cluster persists throughout 2008 and 2009, correctly capturing the prolonged nature of that downturn. The 2020 COVID-19 shock produces a sharp, near-vertical transition into State 1, consistent with the sudden onset of the global liquidity crunch and the subsequent safe-haven rally in gold. The 2022–2024 period presents a more complex pattern: the model identifies intermittent regime switches during the Federal Reserve's quantitative tightening cycle and the inflationary environment, with gold's conditional volatility spiking in tandem with elevated geopolitical tensions.

The estimated transition matrix

$$\hat{A} = \begin{pmatrix} 0.9835 & 0.0165 \\ 0.0331 & 0.9669 \end{pmatrix} \tag{10}$$

confirms the high persistence of both regimes. The probability of remaining in State 0 across consecutive periods is $\hat{A}_{00} = 0.9835$, implying an expected regime duration of approximately $1/(1 - 0.9835) \approx 61$ trading days before a transition to State 1 occurs. Symmetrically, $\hat{A}_{11} = 0.9669$ implies an expected stress regime duration of roughly $1/(1 - 0.9669) \approx 30$ trading days. The low off-diagonal entries ($\hat{A}_{01} = 0.0165$, $\hat{A}_{10} = 0.0331$) confirm that regime switches are rare, discrete events rather than frequent oscillations — a property that underpins the model's utility as an early-warning signal and renders its regime labels economically interpretable for portfolio reallocation.

2.4.5 Out-of-Sample Validation

Walk-Forward Out-of-Sample Design. A key methodological concern in regime-detection models is look-ahead bias: fitting an HMM to the full sample and subsequently reading off historical state assignments amounts to using information unavailable at the time of the decision. To replicate a realistic, real-time decision-making process, we adopt a walk-forward out-of-sample (OOS) scheme. The procedure initializes with a training window of five years ($252 \times 5 \approx 1{,}260$ trading days), fits the two-state Gaussian HMM on that window, and generates state predictions for

the following 22 trading days (approximately one calendar month); these 22 days constitute the genuinely out-of-sample forecast horizon. The window then rolls forward by 22 days, the model is re-estimated on all data available up to the new endpoint, and the process repeats until the full sample is exhausted. At no point does the estimation window incorporate future observations, ensuring that every predicted state label is produced under the same information constraint faced by a real-time practitioner.

OOS Statistical Summary. To verify that the two OOS-predicted states correspond to economically distinct market conditions, we report, for each state, the mean daily log return and standard deviation of both the S&P 500 and gold, together with their pairwise correlation coefficient (Table 4).

Table 4: Out-of-Sample Summary Statistics by Predicted HMM State

	S&P 500		Gold	
Predicted State	Mean (%)	Std Dev (%)	Mean (%)	Corr(r_g, r_{sp})
State 0 (Bull / Calm)	$+0.059$	1.01	$+0.017$	-0.002
State 1 (Bear / Stress)	$+0.033$	1.19	$+0.050$	$+0.070$

Note. Statistics are computed on the out-of-sample predicted state labels produced by the walk-forward scheme. Mean and standard deviation are expressed as daily log returns in percentage terms. Corr(r_g, r_{sp}) denotes the Pearson correlation between gold and S&P 500 log returns within each predicted state.

The conditional means and volatilities reported here are computed on the genuinely out-of-sample prediction window (2011–2026) and therefore differ from the in-sample emission parameters in Table 8 (Appendix B), which are estimated on the full 2006–2026 sample via Baum–Welch.

The directional pattern is consistent across both tables — State 1 is associated with lower equity returns and higher volatility relative to State 0 — but the magnitudes differ, as the OOS figures reflect realized returns under walk-forward predicted labels rather than smoothed in-sample state assignments. The OOS results confirm a statistically meaningful separation between the two regimes. Under State 0, S&P 500 returns exhibit a higher daily mean ($+0.059\%$) and lower volatility (1.01%), consistent with a calm, bull-market environment. Under State 1, mean equity returns compress to $+0.033\%$ while volatility rises to 1.19%, reflecting elevated uncertainty. Notably, gold's mean daily return is *higher* in the stress regime ($+0.050\%$ versus $+0.017\%$), providing preliminary evidence of safe-haven demand during State 1. The near-zero correlation in State 0 (-0.002) and the mild positive correlation in State 1 ($+0.070$) suggest that the two assets do not exhibit the strong negative co-movement traditionally assumed of a safe haven under all conditions; rather, gold's protective role manifests through its *absolute return* during stress, rather than through a sign reversal in its correlation with equities — a finding consistent with the conditional safe-haven hypothesis discussed in Appendix A.

Economic Value Test. To assess whether the statistical regime separation translates into tangible investment value, we construct and compare three strategies over the OOS period, illustrated in Figure 6. The **HMM Strategy** (blue) switches daily between S&P 500 and gold according to the

walk-forward OOS state prediction: hold S&P 500 when State 0 is predicted, and rotate into gold when State 1 is predicted. The **S&P 500 Buy-and-Hold** (orange) serves as the passive benchmark, maintaining full equity exposure throughout. The **VIX $<$ 30 Strategy** (green) provides a rule-based comparison: it holds S&P 500 when the prior day's VIX falls below 30, and switches to gold otherwise; the one-day lag eliminates any look-ahead bias. All three strategies are initialized at a normalized value of 1.0 at the start of the OOS period.

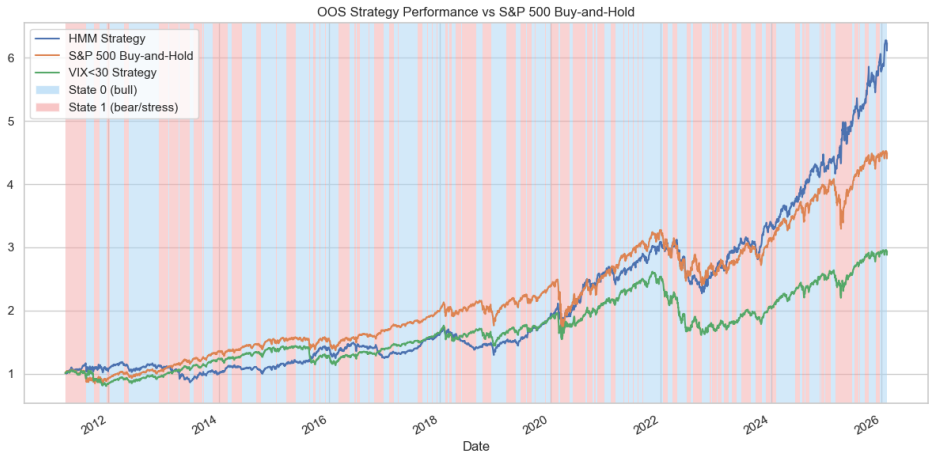

Figure 6: OOS cumulative performance of the HMM Strategy, S&P 500 Buy-and-Hold, and VIX $<$ 30 Strategy (2011–2026). Blue shading denotes State 0 (bull/calm) periods; red shading denotes State 1 (bear/stress) periods as identified by the walk-forward HMM.

The HMM Strategy achieves a terminal value of approximately $6.3\times$ the initial investment, meaningfully outperforming both the S&P 500 Buy-and-Hold ($\approx 4.5\times$) and the VIX $<$ 30 Strategy ($\approx 3.0\times$) over the same horizon. The outperformance is most pronounced during the 2020 COVID-19 shock and the 2022–2024 stress period, where the model's timely rotation into gold preserves capital during equity drawdowns. The VIX $<$ 30 Strategy, by contrast, lags persistently behind the passive benchmark. The underperformance of the VIX threshold strategy relative to the passive benchmark reflects the well-documented cost of binary switching rules under noisy signals (Carr, 2019); this renders it a conservative rather than strong baseline for comparison.

3 Conclusion, Limitation and Future Directions

3.1 Conclusion

This paper examines gold's conditional safe-haven properties over 2006–2026 through a three-stage pipeline: GARCH-X spillover modeling, quantile regression, and a Gaussian Hidden Markov Model.

Three principal findings emerge. The GARCH-X spillover coefficients ($\hat{\gamma} = 0.380$; $\hat{\gamma}_1 = 0.401$, $\hat{\gamma}_2 = 0.364$) confirm that equity stress transmits into gold's conditional variance symmetrically across up- and down-moves, indicating that gold is not volatility-insulated from systemic shocks. Quantile regression reveals that gold's S&P 500 sensitivity compresses from 0.027 at the median to 0.010 at the lower tail ($\tau = 0.05$), while the VIX coefficient reverses sign from -0.043

to $+0.056$ across tails — evidence that safe-haven activation is a distributional tail phenomenon rather than an average-return property. The walk-forward GARCH-HMM evaluation shows gold's mean daily return rising from $+0.017\%$ in State 0 to $+0.050\%$ in State 1, and the HMM switching strategy achieves a $6.3\times$ terminal value versus $4.5\times$ for buy-and-hold and $3.0\times$ for the VIX < 30 rule.

Collectively, these results support the *conditional safe-haven hypothesis* and demonstrate that a data-driven, multivariate regime classifier offers earlier onset detection and fewer false positives than fixed VIX thresholds.

3.2 Limitations

Data Scope. The analysis relies on U.S.-centric data — S&P 500 as equity benchmark, dollar-denominated gold futures, and FRED macroeconomic variables — which may not capture gold's safe-haven dynamics in European, Asian, or emerging markets where central bank demand plays a larger role (Baur and McDermott, 2010).

GARCH-HMM Assumptions. The two-state Gaussian HMM understates tail mass (excess kurtosis 6.74 persists after EGARCH filtering; Table 11, Appendix C), and restricts the latent state space in ways that may miss intermediate regimes. A multivariate Student-t emission and formal likelihood-ratio tests across state-space dimensionalities are natural extensions.

Look-Ahead in GARCH-X. The spillover coefficients $\hat{\gamma}$ are estimated over the full in-sample period, unlike the walk-forward OOS design applied to the HMM, so their magnitudes may overstate real-time predictive strength.

Structural Change. The baseline GARCH-X and quantile regression models are estimated without structural-break tests despite the sample spanning multiple distinct monetary regimes; time-varying parameter models or Chow tests would provide a more robust account of how gold's safe-haven mechanism has evolved.

Transaction Costs. The OOS strategy evaluation omits transaction costs, bid-ask spreads, and slippage, which may be non-trivial during the higher-frequency regime switches of the 2022–2024 period.

3.3 Future Directions

Several extensions of the current framework represent promising avenues for future research.

Richer Emission Distributions. Replacing the Gaussian emission in the HMM with a multivariate Student-t or skew-t distribution would better accommodate the excess kurtosis documented in Section 2.2 and potentially sharpen state separation during tail events. A joint GARCH-HMM specification — in which emission parameters themselves evolve according to a GARCH process — would further allow the model to capture within-state volatility clustering without aggregating it into state transitions.

Sensitivity Analysis. The current framework fixes several modeling choices that warrant systematic evaluation: the number of HMM hidden states, the VIX threshold used as a benchmark, the walk-forward training window length, and the EGARCH feature set fed into the HMM. A sensitivity analysis varying these parameters would clarify how robust the regime classifications and OOS performance conclusions are to specification decisions.

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A Literature Review

A.1 Gold as a Financial Asset

Gold's Dual Properties
Gold is a unique financial asset characterized by its dual nature: it possesses both commodity and monetary attributes. On the commodity side, gold is widely used in jewelry, technology, and limited industrial applications. On the monetary side, it has historically functioned as a store of value, particularly during periods of financial instability, inflation, or geopolitical uncertainty.

Drivers of Gold Price
The price of gold is influenced by several macro-financial factors, including real interest rates, the strength of the U.S. dollar, and geopolitical risk premiums. A decline in real interest rates reduces the opportunity cost of holding gold, thereby supporting its price. Similarly, because gold is denominated in U.S. dollars, a weaker dollar generally increases global demand for gold.

Compared to Other Precious Metals
Compared to other precious metals such as silver, platinum, and copper, gold is less dependent on industrial demand. Metals like silver and copper tend to behave more like cyclical assets, often moving in tandem with economic activity. In contrast, gold is more strongly linked to monetary demand, central bank reserves, and investor sentiment, which makes it distinct in periods of market stress.

A.2 The S&P 500 as Equity Benchmark

In this study, we use the S&P 500 as the benchmark for equity market performance. The S&P 500 reflects the overall condition of large-cap U.S. equities and serves as a proxy for market risk. The relationship between gold and the S&P 500 is central to evaluating gold's safe-haven properties. If gold exhibits low or negative correlation with equities during market downturns, it may function as a safe-haven asset.

A.3 The Safe-Haven Literature

The modern literature on safe-haven assets is grounded in the distinction between a "hedge" and a "safe haven." Baur and McDermott (2010) define a hedge as an asset that is uncorrelated or negatively correlated with stocks on average, whereas a safe haven is an asset that exhibits such properties specifically during periods of extreme market stress.

Their empirical findings suggest that gold can serve as a hedge in normal market conditions and as a safe haven during extreme downturns, although the safe-haven effect is often short-lived. Baur and McDermott (2010) extend this analysis to international markets and show that the safe-haven property of gold varies across countries and time periods.

Subsequent studies emphasize that gold's safe-haven behavior is state-dependent, meaning that its effectiveness varies across different market regimes. More importantly, recent literature suggests that the safe-haven property is fundamentally a tail phenomenon, activated primarily during

extreme negative market conditions (i.e., the lower tail of the return distribution) (Chernozhukov, 2005).

In addition, empirical evidence from studies on extreme market stress indicates that gold does not always act as a reliable safe haven, particularly in severe liquidity crises where all assets may be sold simultaneously (University of Stirling, 2025). This further highlights the need for a more nuanced, regime-dependent analysis.

A.4 Regime Detection Methods

Identifying market stress regimes is a critical step in evaluating safe-haven behavior. A widely used proxy for market uncertainty is the VIX, which is derived from S&P 500 option prices and reflects expected market volatility. Due to its tendency to spike during market downturns, it is often referred to as a "fear gauge." Carr (2019) explains that this interpretation arises from the pricing of downside protection in options markets, which captures investor expectations of future risk.

Despite its intuitive appeal, the VIX threshold approach (e.g., VIX > 30) has notable limitations. The choice of threshold is somewhat arbitrary, and the signal may lag behind actual market stress (Hannich et al., 2018).

To address these limitations, the literature has introduced several data-driven approaches:

- **Markov Switching Models**, introduced by Hamilton (1989), allow time series to switch between latent regimes such as low and high volatility states.
- **Quantile Regression**, further developed in the context of tail behavior by Chernozhukov (2005), enables analysis across different parts of the return distribution, making it particularly suitable for studying tail dependence.
- **GARCH-family models** are widely used to capture volatility clustering and time-varying variance in financial time series (Engle, 1982; Bollerslev, 1986).
- **Hidden Markov Models (HMM)** extend regime-switching approaches by inferring unobserved states from observed data, without requiring externally defined thresholds (Naujoks et al., 2020).

Taken together, these methods capture different aspects of financial dynamics. GARCH models effectively describe volatility persistence but cannot fully explain tail behavior. Quantile regression provides insights into extreme outcomes but lacks explicit regime identification. HMM enables dynamic state detection but depends on informative input features. Therefore, no single method is sufficient to fully characterize gold's safe-haven mechanism.

B Additional Figures and Tables

Table 5: Discriminative Scores of HMM Input Features

Feature	Description	Discriminative Score
`sp500_egarch_vol`	S&P 500 EGARCH conditional volatility	0.7742
`gold_egarch_vol`	Gold EGARCH conditional volatility	0.7458
`diff_DFII10`	10-yr TIPS yield (first difference)	0.1128
`d_gold_copper`	Gold/copper ratio (first difference)	0.0872
`sp500_log_ret`	S&P 500 log return	0.0656
`gold_log_ret`	Gold log return	0.0019

Note. Discriminative scores measure each feature's contribution to separating the two HMM hidden states; higher values indicate greater regime-discriminating power. Features are sorted in descending order.

Table 6: Regime-Stratified Descriptive Statistics by Asset (2006–2026)

Asset	**Regime**	**Days**	**Return (Ann.)**	**Volatility (Ann.)**	**Skewness**	**Kurtosis (Exc.)**
Gold	Normal (VIX < 30)	4585	10.64%	16.78%	-0.9349	8.4129
	Extreme (VIX > 30)	447	14.21%	29.88%	0.3104	1.8273
	Overall	5032	10.96%	18.32%	-0.5291	7.3230
S&P 500	Normal (VIX < 30)	4585	18.69%	14.15%	-0.1931	2.1110
	Extreme (VIX > 30)	447	-93.51%	46.92%	0.0427	1.8861
	Overall	5032	8.69%	19.55%	-0.4777	12.8437
Copper	Normal (VIX < 30)	4585	14.52%	25.44%	-0.6920	15.8434
	Extreme (VIX > 30)	447	-90.88%	47.98%	-0.0182	1.3026
	Overall	5032	5.14%	28.24%	-0.5879	11.5028
Silver	Normal (VIX < 30)	4585	14.55%	32.50%	-1.8466	29.1774
	Extreme (VIX > 30)	447	-31.90%	54.95%	-0.5904	3.4590
	Overall	5032	10.41%	35.09%	-1.5697	22.0227
Palladium	Normal (VIX < 30)	4585	17.72%	34.48%	-0.3825	5.2744
	Extreme (VIX > 30)	447	-99.20%	59.66%	-0.3689	7.7077
	Overall	5032	7.15%	37.50%	-0.4855	8.6083
Platinum	Normal (VIX < 30)	4585	8.95%	25.98%	-0.8738	11.5033
	Extreme (VIX > 30)	447	-85.84%	41.14%	-0.5104	3.6132
	Overall	5032	2.16%	27.38%	-0.8750	10.3799

Note. Returns and volatilities are annualized from daily log returns. Kurtosis is reported in excess of 3. Regime classification follows Carr (2019): Normal = VIX < 30; Extreme = VIX ≥ 30.

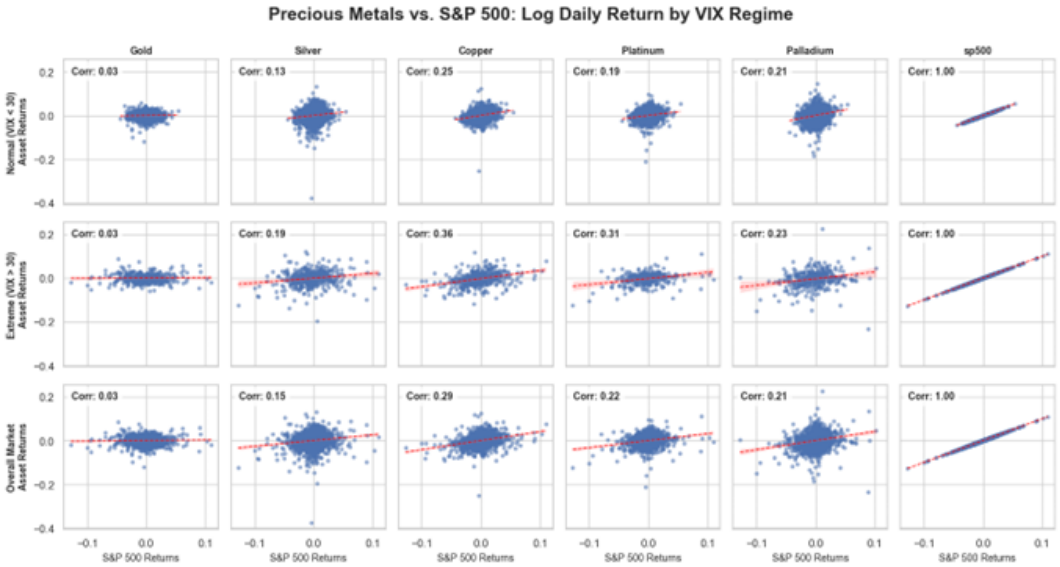

Figure 7: Precious Metals vs. S&P 500: Log Daily Return by VIX Regime. Each panel reports the Pearson correlation coefficient. Rows correspond to VIX regimes (Normal, Extreme, Overall); columns to individual assets.

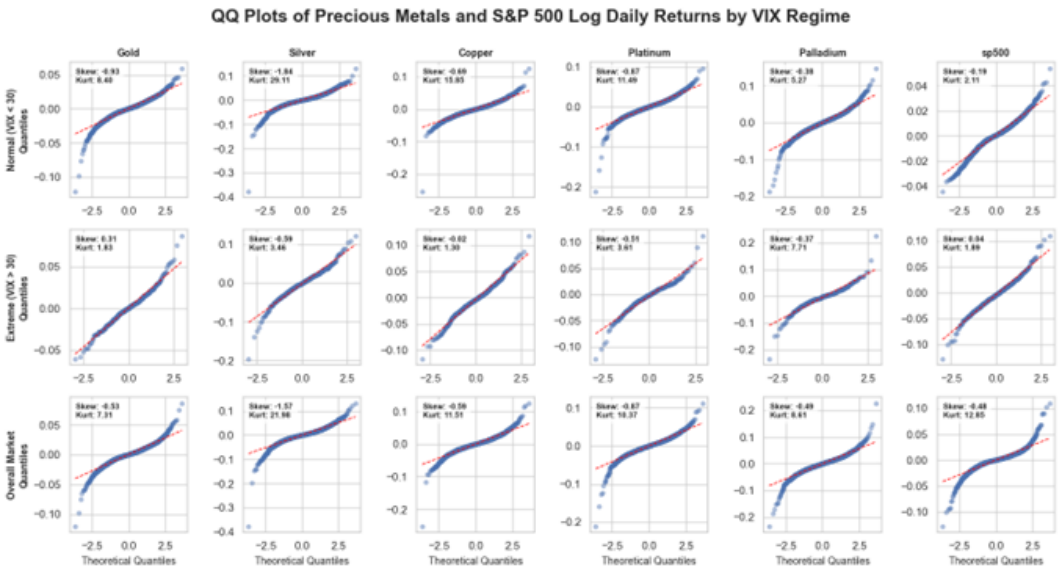

Figure 8: QQ Plots of Precious Metals and S&P 500 Log Daily Returns by VIX Regime. Deviation from the diagonal indicates departure from normality; skewness and excess kurtosis are reported in each panel.

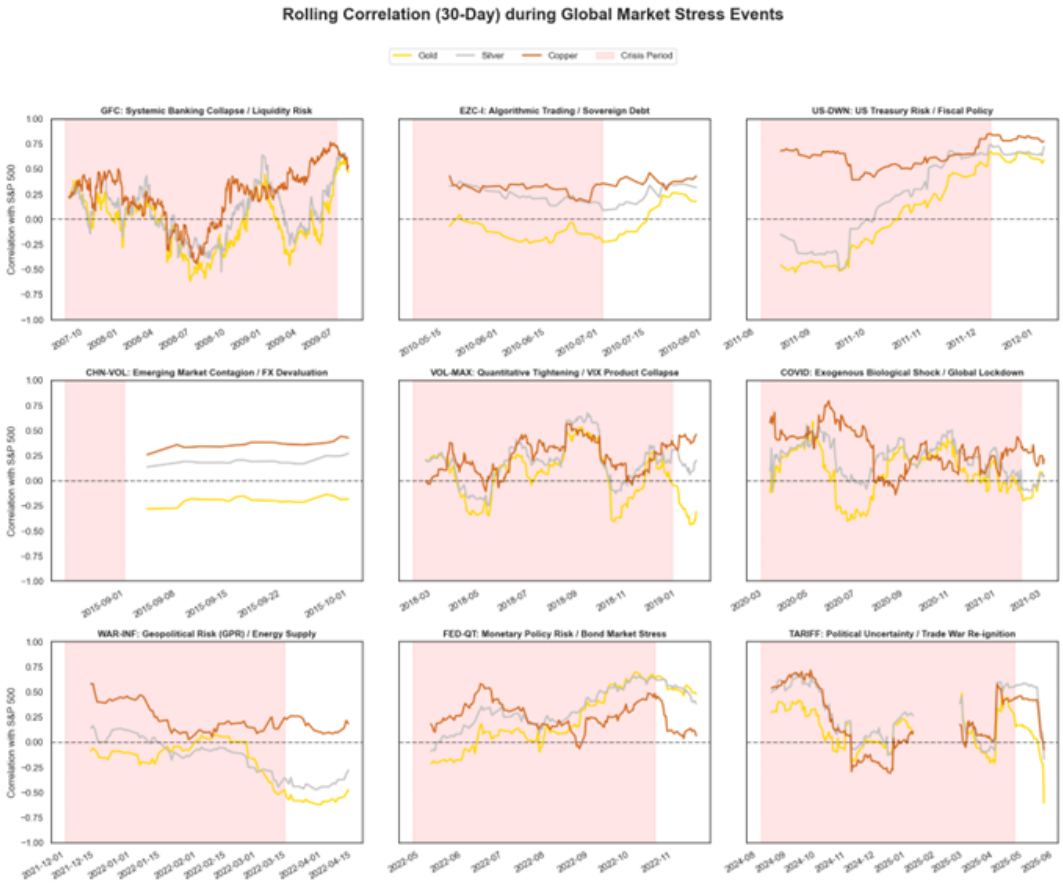

Figure 9: Rolling 30-Day Correlation with S&P 500 during Nine Global Market Stress Episodes (2006–2026). Pink shading denotes the stress window for each episode. Series shown: Gold (yellow), Silver (grey), Copper (brown).

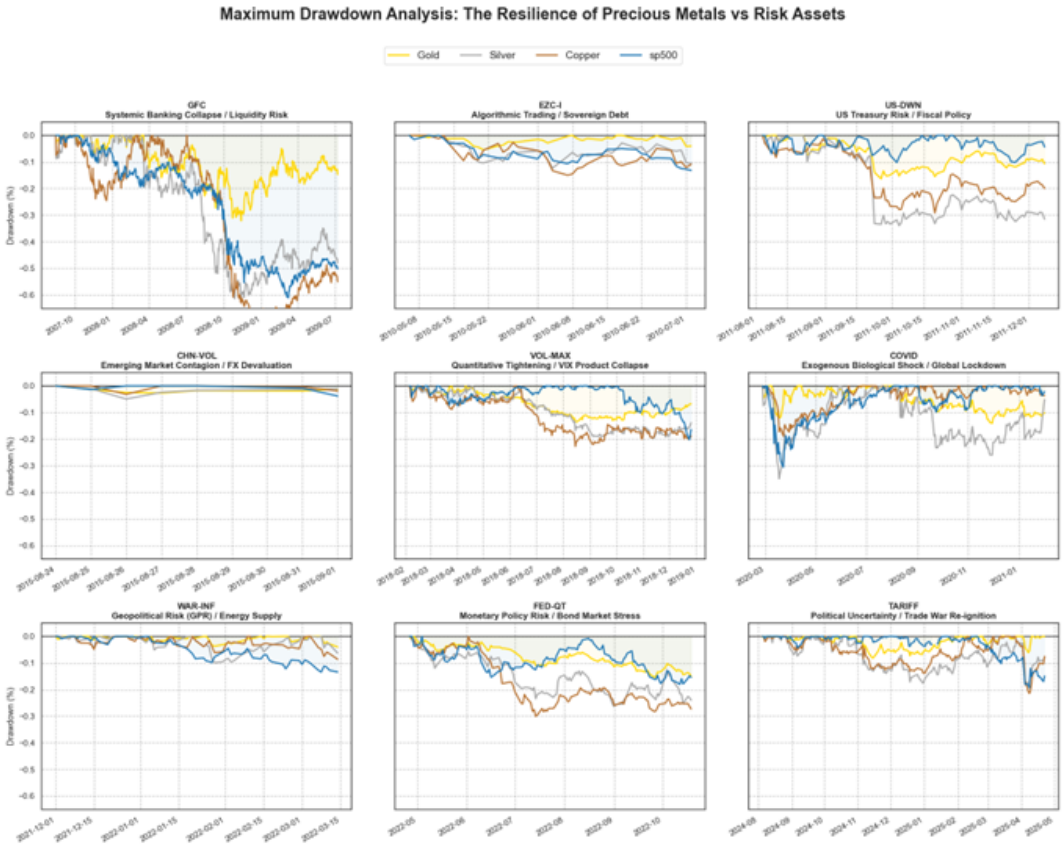

Figure 10: Maximum Drawdown Analysis across Nine Stress Episodes (2006–2026). Series shown: Gold (yellow), Silver (grey), Copper (brown), S&P 500 (blue). Drawdown is expressed as percentage decline from the rolling peak.

Table 7: Estimated Covariance Matrices by HMM Hidden State

Panel A: State 0 (Low-Volatility / Calm Regime)

	r_{gold}	r_{sp}	ΔTIPS	ΔGC ratio	σ^{gold}	σ^{sp}
r_{gold}	0.5482	0.0237	−0.1613	0.1795	0.0090	0.0022
r_{sp}	0.0237	0.3294	0.0332	−0.1094	0.0098	0.0298
ΔTIPS	−0.1613	0.0332	0.6361	−0.1223	0.0054	0.0010
ΔGC ratio	0.1795	−0.1094	−0.1223	0.5610	−0.0154	−0.0149
σ^{gold}	0.0090	0.0098	0.0054	−0.0154	0.2825	0.0286
σ^{sp}	0.0022	0.0298	0.0010	−0.0149	0.0286	0.1478

Panel B: State 1 (High-Volatility / Stress Regime)

	r_{gold}	r_{sp}	ΔTIPS	ΔGC ratio	σ^{gold}	σ^{sp}
r_{gold}	1.9079	0.0364	−0.2990	0.5665	−0.0149	0.0207
r_{sp}	0.0364	2.3350	0.2642	−0.5736	0.0763	0.0955
ΔTIPS	−0.2990	0.2642	1.7313	−0.1993	0.0110	0.1113
ΔGC ratio	0.5665	−0.5736	−0.1993	1.8762	0.0131	0.1373
σ^{gold}	−0.0149	0.0763	0.0110	0.0131	1.3676	0.4437
σ^{sp}	0.0207	0.0955	0.1113	0.1373	0.4437	1.6157

Note. Features: r_{gold} = gold log return; r_{sp} = S&P 500 log return; ΔTIPS = first difference of 10-year TIPS yield (`DFII10`); ΔGC ratio = first difference of gold/copper ratio; σ^{gold} = gold EGARCH conditional volatility; σ^{sp} = S&P 500 EGARCH conditional volatility. States are identified by a 2-state Gaussian HMM fitted to the six-dimensional observation vector. State 0 exhibits uniformly lower diagonal variances (calm regime); State 1 exhibits markedly higher variances across all features (stress regime).

Table 8: HMM State Conditional Statistics (2006–2026)

Parameters	**State 0 (Normal)**	**State 1 (Crisis)**
S&P 500 Mean Return (%)	+0.091	−0.078
Gold Mean Return (%)	+0.046	+0.041
S&P 500 Volatility (%)	0.777	1.504
Gold Volatility (%)	0.958	1.375

Note. Statistics are derived from the in-sample Viterbi-decoded state sequence over the full estimation period (2006–2026). All return and volatility figures are expressed as daily log returns in percentage terms.

C GARCH-X Residual Diagnostics

Table 9 and Table 10 report the parameter estimates and spillover test results for the GARCH-X baseline. Table 11 and Figure 11 report the distributional diagnostics for the standardized residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$ from the symmetric and asymmetric GARCH-X specifications. Both the numerical summary and the visual inspection confirm that volatility filtering removes clustering but leaves substantial tail mass unabsorbed, motivating the regime-switching extension in Section 2.4.

Table 9: Symmetric and Asymmetric GARCH Baseline Summary

	Symmetric			*Asymmetric*		
Parameter	Coef.	Std. Err.	p-value	Coef.	Std. Err.	p-value
Mean Model						
μ	0.0601	1.226e-02	9.647e-07	0.0601	1.226e-02	9.647e-07
Volatility Model						
ω	8.3992e-03	2.948e-03	4.381e-03	8.3992e-03	2.948e-03	4.381e-03
α	0.0387	5.852e-03	3.670e-11	0.0387	5.852e-03	3.670e-11
β	0.9569	6.528e-03	0.000	0.9569	6.528e-03	0.000
Distribution						
ν	4.4897	0.290	4.754e-54	4.4897	0.290	4.754e-54
Model Fit						
Log-Likelihood		-7150.23			-7150.23	
AIC		14310.5			14310.5	
BIC		14343.1			14343.1	
Observations		5012			5012	

Vol Model: GARCH. Distribution: Standardized Student's t. Method: Maximum Likelihood.

Table 10: Symmetric and Asymmetric Spillover γ Tests (OLS, HC3 Robust Standard Errors)

	Coef.	Std. Err.	z-stat	p-value
Panel A: Symmetric Spillover				
Intercept	1.0786	0.021	52.609	0.000
$\lvert r_{\text{sp500},t-1}\rvert$	0.3801	0.026	14.594	0.000
R^2	0.116			
F-statistic	213.0	$(p = 2.81 \times 10^{-47})$		
Observations	5012			
Panel B: Asymmetric Spillover				
Intercept	1.0761	0.020	52.878	0.000
$r^{+}_{\text{sp500},t-1}$	0.4014	0.031	12.842	0.000
$r^{-}_{\text{sp500},t-1}$	0.3637	0.033	11.108	0.000
R^2	0.116			
F-statistic	111.7	$(p = 3.42 \times 10^{-48})$		
Observations	5012			

Dependent variable: $\hat{\sigma}_t^2$ (Gold conditional variance). Covariance type: HC3.
r^+ and r^- denote lagged positive and lagged negative S&P 500 returns, respectively.

Table 11: Distributional Gap Report for Standardized Residuals

Model	N	Mean	Std. Dev.	Kurtosis	JB Stat	JB p-value
Symmetric GARCH	5012	-0.0171	0.9887	6.7393	3118.18	≈ 0
Asymmetric GARCH	5012	-0.0171	0.9887	6.7393	3118.18	≈ 0

Mean and Std. Dev. refer to the standardized residuals $\hat{z}_t = \hat{\varepsilon}_t/\hat{\sigma}_t$. Kurtosis benchmark under normality is 3. JB: Jarque–Bera test of normality.

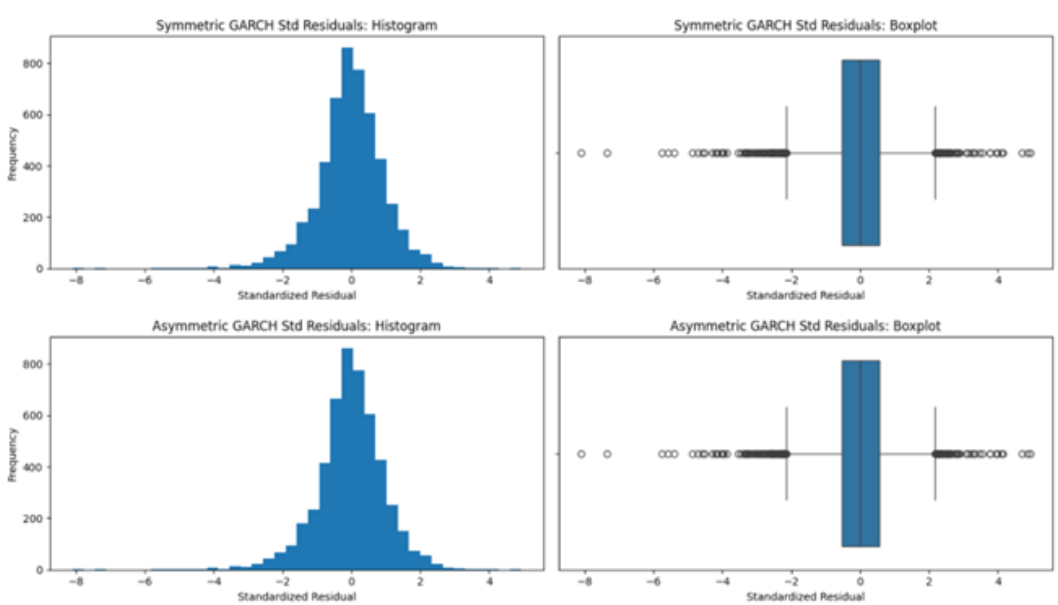

Figure 11: Standardized residual histograms and boxplots for the symmetric and asymmetric GARCH-X baselines.

D Quantile Regression Full Coefficient Table

Table 12 reports the full set of quantile regression coefficient estimates for gold log returns on S&P 500 returns and VIX across seven quantile levels, together with 95% confidence intervals and Koenker–Bassett pseudo-R^2.

Table 12: Quantile Regression Coefficients (Full)

τ	$\hat{\alpha}$	$\hat{\beta}_{\text{sp500}}$	$\hat{\beta}_{\text{vix}}$	$\text{CI}^{\text{sp500}}_{\text{low}}$	$\text{CI}^{\text{sp500}}_{\text{high}}$	$\text{CI}^{\text{vix}}_{\text{low}}$	$\text{CI}^{\text{vix}}_{\text{high}}$	Pseudo-R^2
0.05	−0.0097	0.0099	−0.0426	−0.0853	0.1051	−0.0547	−0.0306	0.0414
0.10	−0.0050	0.0304	−0.0387	−0.0273	0.0881	−0.0464	−0.0309	0.0352
0.25	−0.0014	0.0424	−0.0187	0.0131	0.0717	−0.0229	−0.0146	0.0102
0.50	0.0001	0.0266	0.0021	0.0043	0.0490	−0.0011	0.0052	0.0004
0.75	0.0025	0.0332	0.0211	0.0029	0.0635	0.0169	0.0253	0.0105
0.90	0.0052	0.0366	0.0408	−0.0121	0.0853	0.0347	0.0469	0.0386
0.95	0.0068	0.0190	0.0555	−0.0594	0.0975	0.0458	0.0652	0.0565

τ: quantile level. $\hat{\beta}_{\text{sp500}}$ and $\hat{\beta}_{\text{vix}}$: slope coefficients on S&P 500 returns and VIX, respectively. CI_{low} and CI_{high}: 95% confidence interval bounds. Pseudo-R^2: Koenker–Bassett goodness-of-fit measure.